

Branching explains why calculation needs focus

Chess Math

THE IDEA

If you examine three candidates and three replies to each, you already have nine two-ply branches. Adding three continuations per reply produces 27 branches. Practical calculation controls this growth by prioritizing forcing moves and critical defenses.

WORKED EXAMPLE

A line with a forced king reply is easier to visualize than one where the defender has ten quiet choices. This does not make the forcing line best, but it gives you a sensible place to begin your search.

TRY IT BEFORE READING ON

How many leaf positions arise from three choices at each of four plies?

ANSWER AND REASONING

$3 \times 3 \times 3 \times 3 = 81$. Real chess branching is uneven, so do not calculate every branch equally. Eliminate unsound candidates early and spend effort on plausible resistance.

TAKE THIS INTO YOUR NEXT GAME

Use candidate selection to spend calculation where it matters.